Computer Architecture

Homework 1

1)a)addi x8, x5, -5

Explanation: subi doesn't exist in RISC-V, so we use addi with a negative immediate value to achieve the same result.

b)slli x5, x3, 3

Explanation: Instead of using mul which is a costlier operation, we use left shift by 3 which is equivalent to multiplying the number by 8.

c)add x19, x19, x10

Explanation: x19 += x10 is equivalent to x19 = x19 + x10 and hence we use the add instruction with x19 appearing twice so we can assign the value stored in x19 as x19 + x10.

d)addi x15, x15, 1

Explanation: ++x15 is equivalent to x15 = x15 + 1.

e)srli x19, x15, 2

Explanation: divi does not exist in the RISC-V ISA and hence we use the srli instruction to shift the bits of the number stored in x15 to the right by 2 which is equivalent to dividing the number stored in x15 by 4.

f)addi x12, x19, 4

Explanation: Adding a variable value in a register to a fixed value (immediate) requires us to use addi.

2)a)ld x6, 160(x5)

addi x6, x6, 100

sd x6, 96(x5)

b)ld x6, 160(x5)

addi x6, x6, 1

sd x6, 160(x5)

c)ld x6, 40(x5)

ld x7, 96(x5)

sd x7, 40(x5)

sd x6, 96(x5)

d)ld x6, 32(x5)

slli x7, x6, 32

srli x7, x7, 32

sd x7, 32(x5)

e)ld x6, 16(x5)

slli x7, x6, 32

srli x8, x6, 32

or x9, x7, x8

sd x9, 16(x5)

3)a)$+23_{10} = 0001\ 0111_2$

+23 = 11*2 + 1

11 = 5*2 + 1

5 = 2*2 + 1

2 = 2*1 + 0

Therefore, $23_{10} = 0001\ 0111_2$.

b)$-1_{10} = 1111\ 1111_2$

+1 = 0000 0001

-1 = Complement of(+1) + 1 (0000 0001) = 1111 1110 + 0000 0001 = 1111 1111

c)$+255_{10}$: Cannot be represented in 2's complement form for 8 bits as the range of integers that can be represented in this form for 8 bits is -2^8 to $2^8 - 1$ or -128 to +127 which does not include +255.

d)-128_{10}: $1000\ 0000_2$

$+128 = 2^7 = 1000\ 0000$ (+128 cannot be represented in 8 bits signed form and hence we are using the unsigned form representation for representing +128 in binary.)

-128_{10}: 0111 1111 + 0000 0001 = $1000\ 0000_2$ which is lowest possible value representable in the 2's complement form for 8 bits.

4)a)1101 0100 has Most Significant Bit(MSB) = 1 and it therefore represents a negative value in 2's complement representation. Complement of 1101 0100 = 0010 1011 = $2^0 + 2^1 + 2^3 + 2^5 = 43$ represents a positive integer.

Therefore, 1 + 1101 0100 + 0010 1011 = 1 0000 0000 = 0 in 2's complement for 8 bits.

Therefore, 1101 0100 + 1 = -0010 1011 in 2's complement for 8 bits.

Therefore, $1101\ 0100_2 = -44_{10}$.

b)0010 1011 represents a positive integer.

As we have seen above in part a, this number represents 43.

Therefore, $0010\ 1011_2 = 43_{10}$.

c)1111 1110 has Most Significant Bit (MSB) = 1 which implies it represents a negative integer.

Complement of 1111 1110 is 0000 0001.

Therefore, 0000 0001 + 1111 1110 + 1 = 1 0000 0000 = 0 in 2's complement for 8 bits.

Therefore, $1111\ 1110_2 = -2_{10}$.